

Jinhui Yi

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EDUCATION

University of Bonn

Ph.D. in Computer Vision

2020 – present

Bonn, Germany

- Supervisor: Prof. Dr. Juergen Gall | Computer Vision Group
- Topics: Multimodal Video Understanding, Domain Adaptation, Plant Phenotyping
- 15 peer-reviewed publications (7 as first or joint-first author) at venues including CVPR, ICCV, ECCV, ICLR

University of Hannover

M.Sc. in Electrical Engineering and Information Technology

2016 – 2019

Hanover, Germany

- Supervisor: Prof. Dr. Bodo Rosenhahn | Institute of Information Processing
- Specialization: Machine Learning, Computer Vision
- Thesis: Graph-based Deep Reinforcement Learning for Semantic Image Analysis

South China University of Technology

B.Sc. in Electronic Science and Technology

2011 – 2015

Guangzhou, China

- Specialization: Digital Electronics, Communication Technology

WORK EXPERIENCE

Lamarr Institute for Machine Learning and AI

Research Assistant

May 2024 – present

Bonn, Germany

- Led Video-Panda (CVPR 2025), the first encoder-free video-language model, replacing standard 300M to 1.4B parameter visual encoders with a 45M-parameter Spatio-Temporal Alignment Block for a $6.5\times$ parameter reduction and 3 to $4\times$ throughput gain; training scaled across hundreds of GPUs using PyTorch DDP.
- Co-led Open-Vocabulary Long Term Action Anticipation (ECCV 2026), the first open-vocabulary evaluation framework for action anticipation; the proposed horizon-specific learnable queries and lightweight text-encoder adaptation clearly outperform existing anticipation methods.
- Contributing author on RedSage (ICLR 2026), an open-source, locally deployable 8B cybersecurity LLM trained on a self-curated 11.8B-token continual-pretraining corpus and 266K agentic multi-turn SFT samples, surpassing baselines by up to +5.59 points on cybersecurity benchmarks (CTI-Bench, CyberMetric, SECURE) and +5.05 points on the Open LLM Leaderboard.
- Co-supervised student research on multi-modal object and person re-identification, resulting in five further publications at ICASSP, ICME, ICIP, and IJCNN ($\times 2$).

Cluster of Excellence PhenoRob, University of Bonn

Research Assistant

Jan. 2020 – Apr. 2024

Bonn, Germany

- Led the research line on non-invasive nutrient deficiency diagnosis from RGB images, spanning dataset collection, benchmarking, and domain adaptation.
- Proposed MV-Match (BMVC 2024), the first approach to exploit multiple camera views in both the labeled source and the unlabeled target domain for unsupervised domain adaptation, achieving state-of-the-art results on two nutrient deficiency datasets.
- Led the UAV-based study published in Computers and Electronics in Agriculture (COMPAG 2025): collected more than 3,600 drone RGB images of winter wheat and winter rye in a long-term fertilizer experiment and benchmarked eight CNN- and transformer-based models.
- Collected and released the Deep Nutrient Deficiency for Sugar Beet (DND-SB) dataset (Sensors 2020), containing 5,648 RGB images covering N, P, K, and lime omission as well as full and zero fertilization, and benchmarked it with CNN-based models to establish baselines and expose the limits of RGB-only diagnosis.

Bosch Center for Artificial Intelligence

Research Intern

Mar. 2018 – Sep. 2018

Renningen, Germany

- Supervisor: Dr. Michael Volpp
- Implemented and evaluated PPO, TRPO, and DDPG in TensorFlow, and ran hyperparameter optimization of deep reinforcement learning baselines on OpenAI Gym.

Institute of Information Processing, University of Hannover

Research Assistant

Nov. 2017 – Mar. 2018

Hanover, Germany

- Supervisor: Dr. Hanno Ackermann
- Reimplemented AlphaZero with Monte Carlo tree search for Gomoku.

SKILLS

Research Areas: Multimodal Video Understanding, Vision-Language Models, Encoder-free Architectures, Long-Term Action Anticipation, Unsupervised Domain Adaptation, Plant Phenotyping
Programming: Python (Advanced)
Deep Learning: PyTorch, HuggingFace Transformers, distributed training (DDP, FSDP, DeepSpeed)
HPC and Tooling: SLURM, Weights & Biases, OpenCV, NumPy, Pandas, Scikit-Learn
Languages: English (C1), German (C1), Chinese (Native), Japanese (Basic)

ACADEMIC SERVICES

Conference / Journal Reviewer: CVPR, ICCV, ECCV, WACV, AAAI, IJCV
Co-organizer: Workshop on Computer Vision in Plant Phenotyping and Agriculture (CVPPA@ICCV'2023); German Conference on Pattern Recognition (GCPR'2021)

AWARDS AND GRANTS

- **Compute Award (Jupiter Supercomputer):** 750K GPU compute hours for Holistic Multi-modal Egocentric Video Forecasting (2026)
- **Compute Award (Leonardo Supercomputer):** 1.0M GPU compute hours across three allocations for Unified Multimodal Video Anticipation and Forecasting (2026), Encoder-Free Video Language Models (2026), and Large-Scale Open-Vocabulary Video Understanding and Anticipation (2025)

PUBLICATIONS

* indicates equal contribution

Conferences

1. S. T. Wasim*, **J. Yi***, H. Suleman, A. Javed, Y. Luo, M. Naseer, and J. Gall, "Open-vocabulary long term action anticipation," in *ECCV*, 2026
2. N. Suryanto, M. Naseer, P. Li, S. T. Wasim, **J. Yi**, J. Gall, P. Ceravolo, and E. Damiani, "RedSage: A cybersecurity generalist LLM," in *ICLR*, 2026
3. X. Ma, **J. Yi**, and J. Gall, "CoRe-Net: Consensus-based selection and reciprocal reliability for multi-modal object re-identification," in *ICIP*, 2026
4. X. Ma, Y. Wang, **J. Yi**, and J. Gall, "EPIC: Semantic inverse prompting and evidential corroboration for multi-modal object re-identification," in *IJCNN*, 2026
5. X. Ma, Y. Wang, **J. Yi**, and J. Gall, "AdapDeFormer: Adaptive deformable attention aggregation with hierarchical query learning for multi-modal object re-identification," in *IJCNN*, 2026
6. X. Ma, Y. Wang, **J. Yi**, R. Zhang, P. Zhang, and J. Gall, "SynergyReID: Bridging modalities via semantic prototypes and consistency priors for multi-modal object re-identification," in *ICME*, 2026
7. **J. Yi***, S. T. Wasim*, Y. Luo*, M. Naseer, and J. Gall, "Video-Panda: Parameter-efficient alignment for encoder-free video-language models," in *CVPR*, 2025
8. X. Ma, **J. Yi**, and J. Gall, "PeT-KeyStAtion: Parameter-efficient transformer with keypoint-guided spatial-temporal aggregation for video-based person re-identification," in *ICASSP*, 2025
9. **J. Yi***, Y. Luo*, M. Deichmann, G. Schaaf, and J. Gall, "MV-Match: Multi-view matching for domain-adaptive identification of plant nutrient deficiencies," in *BMVC*, 2024
10. Y. Luo*, **J. Yi***, Y. A. Farha, M. Wolter, and J. Gall, "Rethinking temporal self-similarity for repetitive action counting," in *ICIP*, 2024
11. Y. Cong, **J. Yi**, B. Rosenhahn, and M. Y. Yang, "SSGVS: Semantic scene graph-to-video synthesis," in *CVPRW*, 2023
12. S. Li, Y. Zhou, **J. Yi**, and J. Gall, "Spatial-temporal consistency network for low-latency trajectory forecasting," in *ICCV*, 2021

Journals

1. **J. Yi**, G. Lopez, S. Hadir, J. Weyler, L. Klingbeil, M. Deichmann, J. Gall, and S. J. Seidel, "Non-invasive diagnosis of nutrient deficiencies in winter wheat and winter rye using UAV-based RGB images," *Computers and Electronics in Agriculture (COMPAG)*, 2025
2. S. Li*, **J. Yi***, Y. A. Farha, and J. Gall, "Pose refinement graph convolutional network for skeleton-based action recognition," *IEEE Robotics and Automation Letters (RA-L)*, 2021
3. **J. Yi**, L. Krusenbaum, P. Unger, H. Hüging, S. J. Seidel, G. Schaaf, and J. Gall, "Deep learning for non-invasive diagnosis of nutrient deficiencies in sugar beet using RGB images," *Sensors*, 2020